The original F4 model has a saturation boundary

Lean proves that every run of the repository's original `F4Run` model stays in the ideal generated by its inputs. It directly disproves the unrestricted claim that every saturation member can appear in such an ordinary run. The original model files are imported byte for byte. [pcc:unverified]

For $I = (xy, x^2 - x, y^2 - y)$ and saturation by x, both y and $x - 1$ have power-1 saturation witnesses. Every generator vanishes at $(0, 1)$ but y does not. Therefore $y \notin I$, and no ordinary model run from these generators can contain the registered saturated rows. The generic proof holds over every field; a separate literal F101 certificate also interprets printed coefficient 100 as -1. [pcc:unverified]

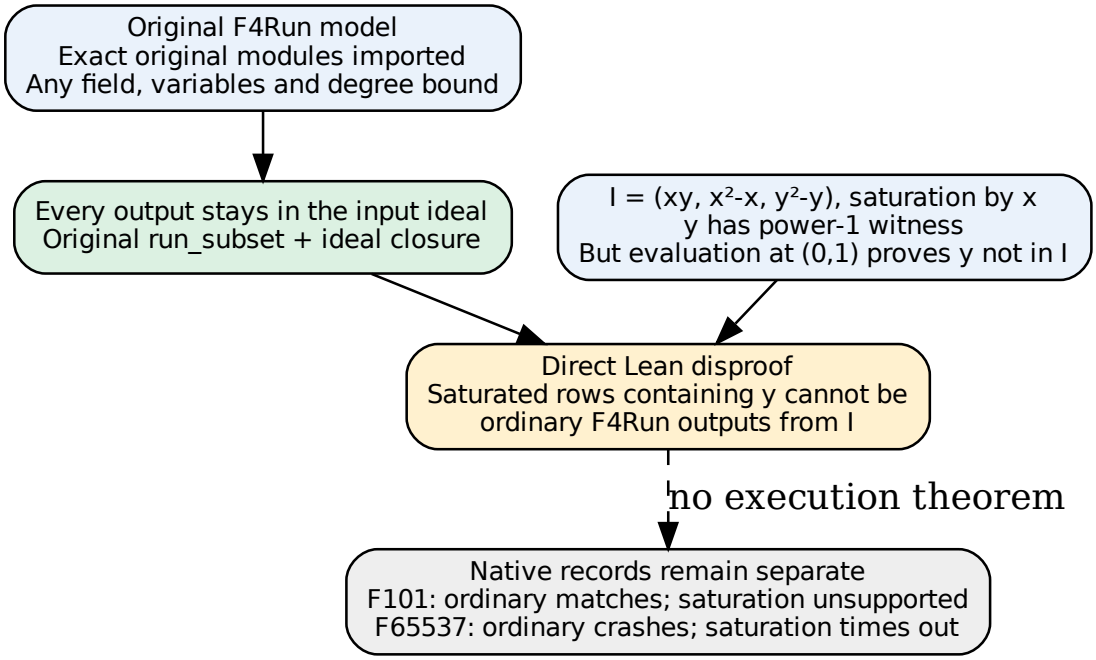


Figure 1: Formal invariant and disproof, separated from native execution. [pcc:unverified]

Native failures and the remaining refinement obligation

Both native protocols fixed inputs, analyzer, binary, expected bases and stopping rules before their affected data, and were committed before calls. All four calls are retained. No retry, parser patch, timing inference or paid compute was used. [pcc:unverified]

Version / field	Ordinary call	Saturation call
v1 / F101	exit 0; expected basis, S-pairs, reductions, staircase 3 and point count 3 match	exit 1; upstream rejects low characteristic; empty basis
v2 / F65537	exit 139; segmentation fault; empty basis	exit 124; 60-second timeout; empty basis

The v1 protocol has an acknowledged registration error: its sample-count field implies four calls while its stopping rule and runner specify two. Its export retains both calls and fails three planning checks. The frozen record is unchanged. V2 explicitly amends the field, sample count, watchdog and checker before its data. Its complete two-row export passes file/schema/ protocol checks, with both solver outcomes `not_validated`. Export validity does not mean solver correctness. No native conformance claim follows. [pcc:unverified]

V2 widens coefficient products to 64 bits and checks only four Boolean candidates; it makes no full-field support-count claim. Synthetic calibration checks a wrong known answer, zero row, field header and maximal coefficient. The checker shares earlier arithmetic and the same host, so it supplies no other-machine independence. The native crash/hang cause remains undiagnosed. [pcc:unverified]

The pinned msolve version is `5e72b9d9` (v0.6.5). Reviewed pair selection and preprocessing construct monomial multiples in C, but a refinement still needs row interpretation, exponent/degree correctness, exact arithmetic and update/trace proofs. The named “simplify” comment occurs in `f4sat.c`, a saturation path. It cannot establish ordinary-model conformance. The imported original Lean module itself describes implementation conformance as a source reading, not its theorem. [pcc:unverified]

Main binding: `F4ExpectedOutputsAudit.expected_output_audit_certificate`. Proof map: `../../F4_MODEL.md`; frozen source plan: `../../f4-model-sources.json`. Whole census gate: 573 theorems, zero failing, zero trusted-native exceptions. Lean 4.35.0 / Mathlib 56a00585; only standard axioms `propext`, `Classical.choice`, `Quot.sound`. No holes, custom axioms, linter waivers or added elaboration option. Failed builds are archived with byte-equal round trips; fresh lean4checker is unavailable. [pcc:unverified]

Theorem packet `PCC-20261110-0f3faf` binds source commit `2ff49e49`. Native artifact packet `PCC-20261110-5286d7` binds both exports; execution is not covered without independent signed evidence. Actual local verdicts and signed standing are recorded separately. Native protocol hashes: v1 `637a5a3b…b6f2`, v2 `e0d22db8…5772`. KB/Conductor unavailable; bounded original-source/Mathlib fallback used. No ordinary C refinement or blanket source promotion. [pcc:unverified]